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Linear Regression with *R*

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Outline

- Basic Concepts
- Model Structure
- Model Results Interpretation
- Model Comparison and Selection
- Model Assumption Checks
- *R* Programming Practical

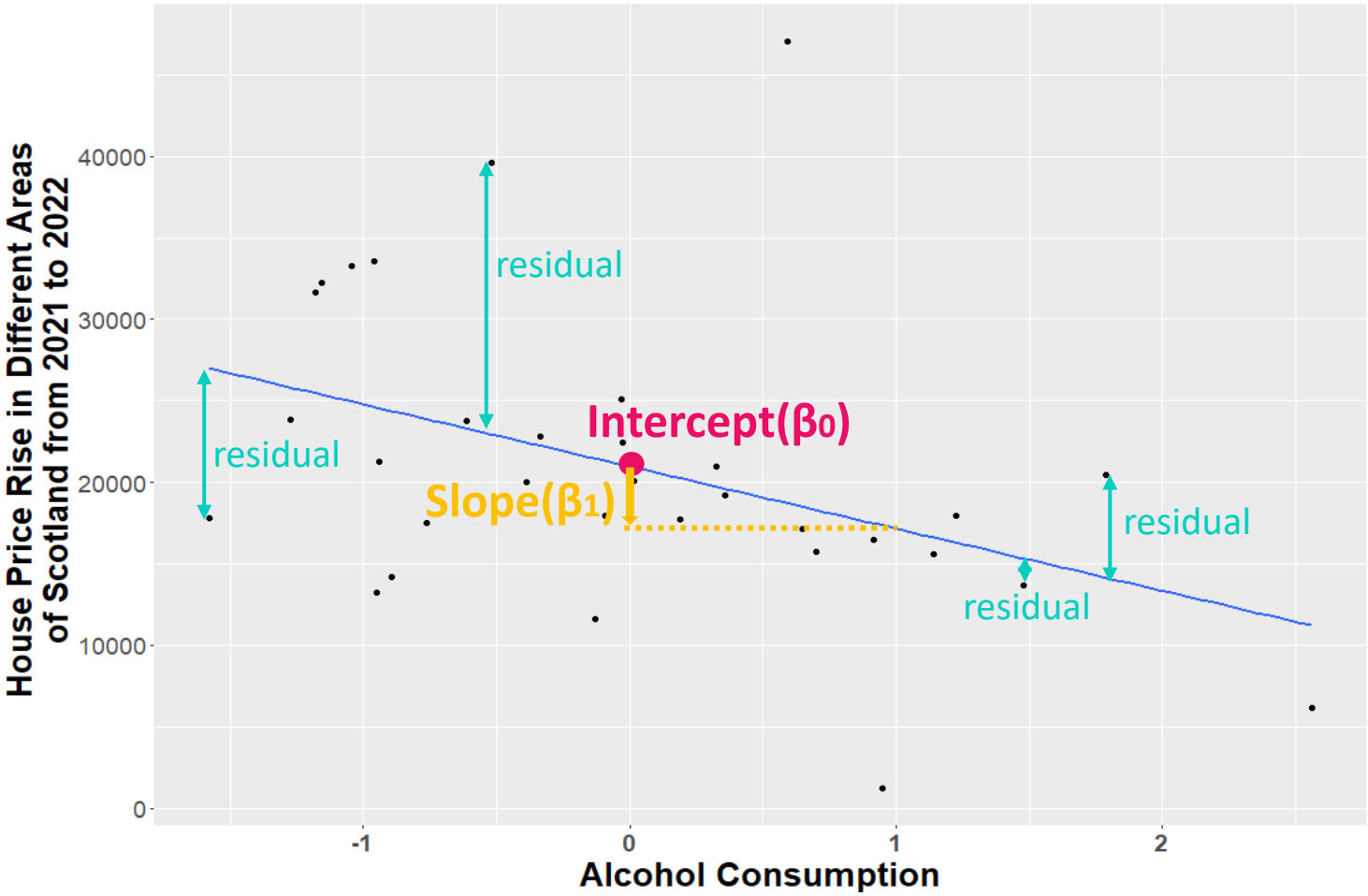
Basic Concepts

Model: a linear relation between two variables x & y

Intercept: the expected value of y when x is 0 or on average (if continuous).

Slope: the rate of change (number of units by which y increases, as x increases by 1 unit).

Residual: the difference between an observed value and the value predicted by the model



Model Structure

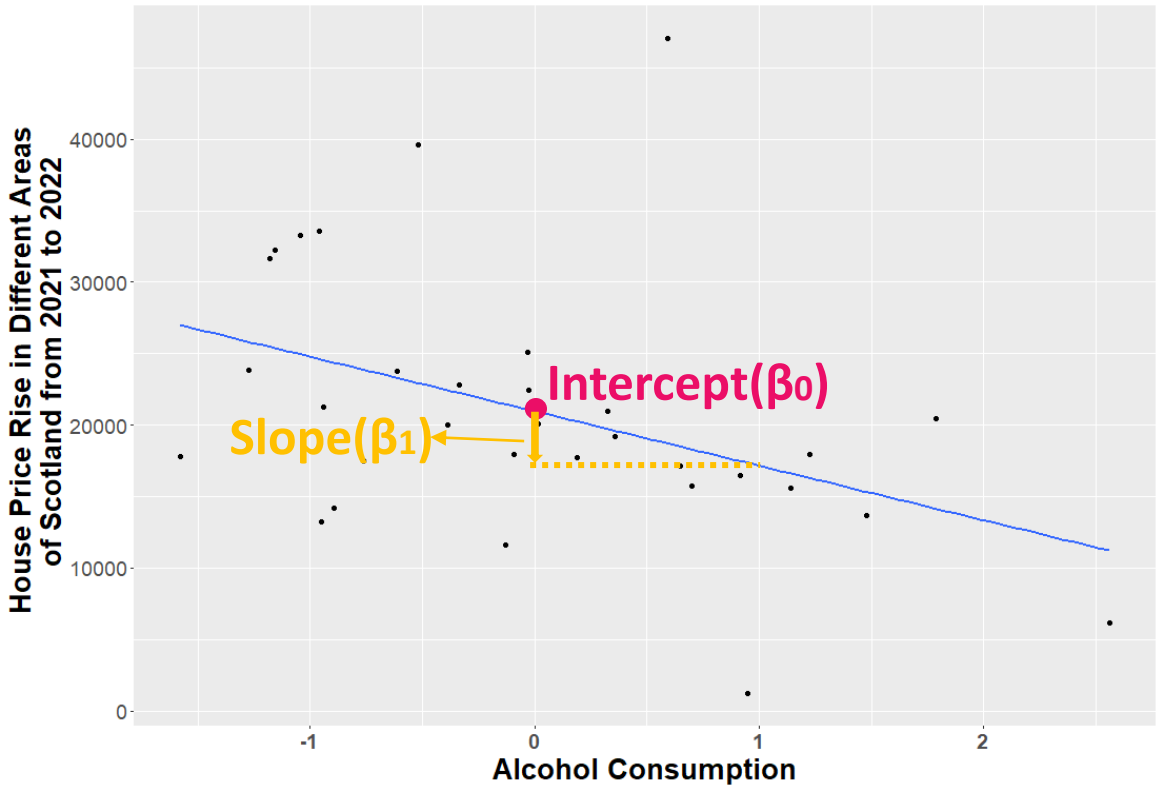
- **Simple linear regression:**

$$Y \sim \beta_0 + \beta_1 X + \varepsilon$$

in R:

lm(Y ~ X, data = data)

lm (house price ~ alcohol consumption,
data = scot)



Multiple linear regression:

- Multiple predictors / independent variables

$$y = \beta 0 + \beta 1\, x1 + \beta 2\, x2 + \cdots + \beta n\, xn + \epsilon$$

- Multiple predictors / independent variables with **interaction**

$$y = \beta 0 + \beta 1\, x1 + \beta 2\, x2 + \beta 3\, x1\, x2 + \cdots + \beta n\, xn + \epsilon$$

- Goal: How each independent variable impacts the dependent variable, while controlling for other variables.

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Model Results Interpretation

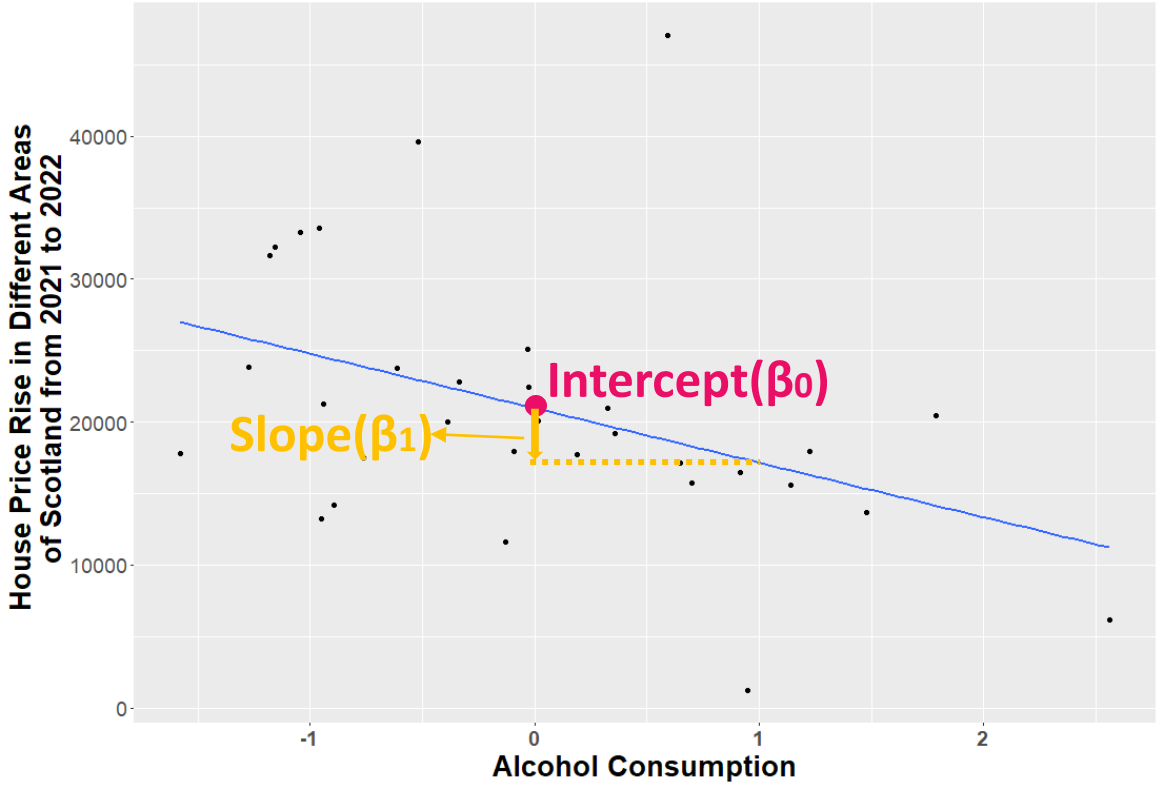
Effect of x on y (slope)

"Y is explained by x."

"Y is predicted by x."

"Y is associated with x."

Can we assume a causal relation ???



What are we testing?

NULL HYPOTHESIS TESTING (NHT)

- Hypotheses are claims researchers make about the world based on some data.
- NHT is a formal method for validating hypotheses, by comparing a theoretically non-null hypothesis (H1) with a null hypothesis (H0).

  H_1: House price rises associated with alcohol consumption.

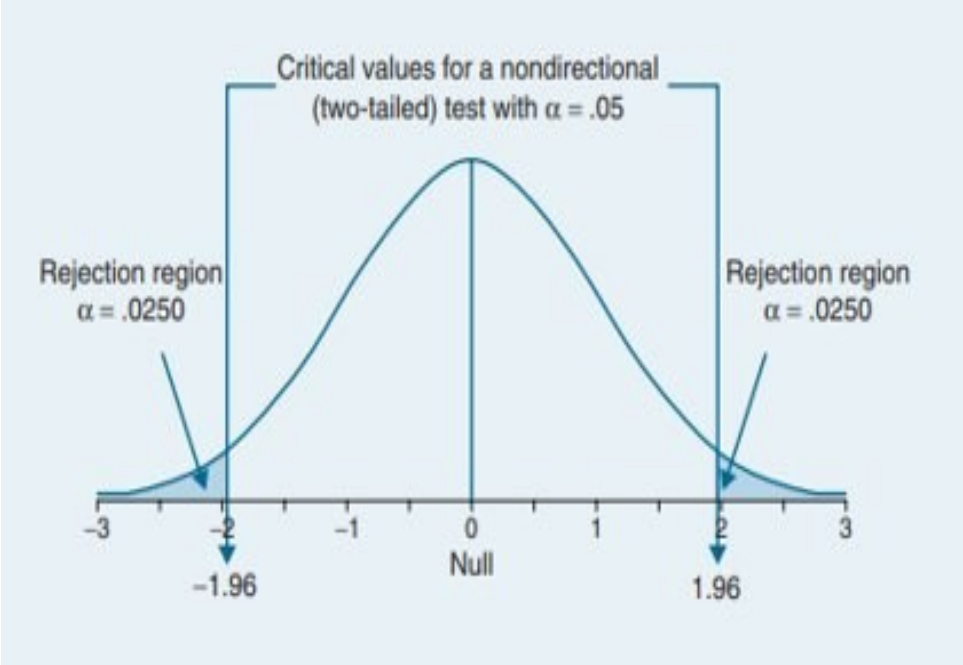
  H_0: House price rises not associated with alcohol consumption.
- The goal is to accept or reject **H0** (not H1) and then make inferences about H1.

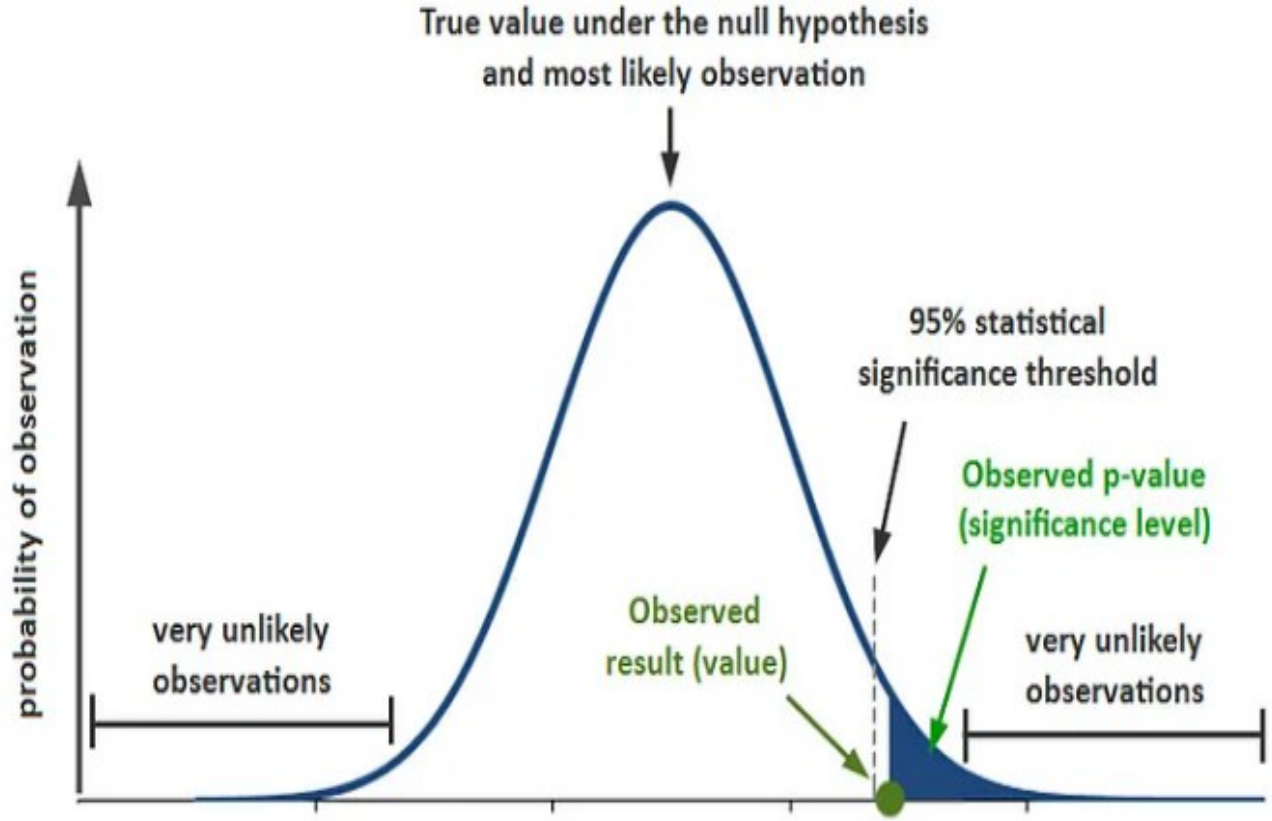
  H_1: All pandas are black and white.

  H_0: Some pandas are not.

NULL HYPOTHESIS TESTING (NHT)

- Goal: Reject or accept H0





Resources: https://medium.com/datadriveninvestor/p-value-t-test-chi-square-test-anova-when-to-use-which-strategy-32907734aa0e; https://www.sagepub.com/sites/default/files/upm-binaries/40007_Chapter8.pdf; https://learningstatisticswithr.com/lsr-0.6.pdf

NULL HYPOTHESIS TESTING (NHT)

- Two types of errors in decision making

		Decision	
		Retain H_0	Reject H_0
H_0 Truth	True	$1 - \alpha$ = probability of correct retention	α (type I error rate)
	False	β (type II error rate)	$1 - \beta$ = power of the test

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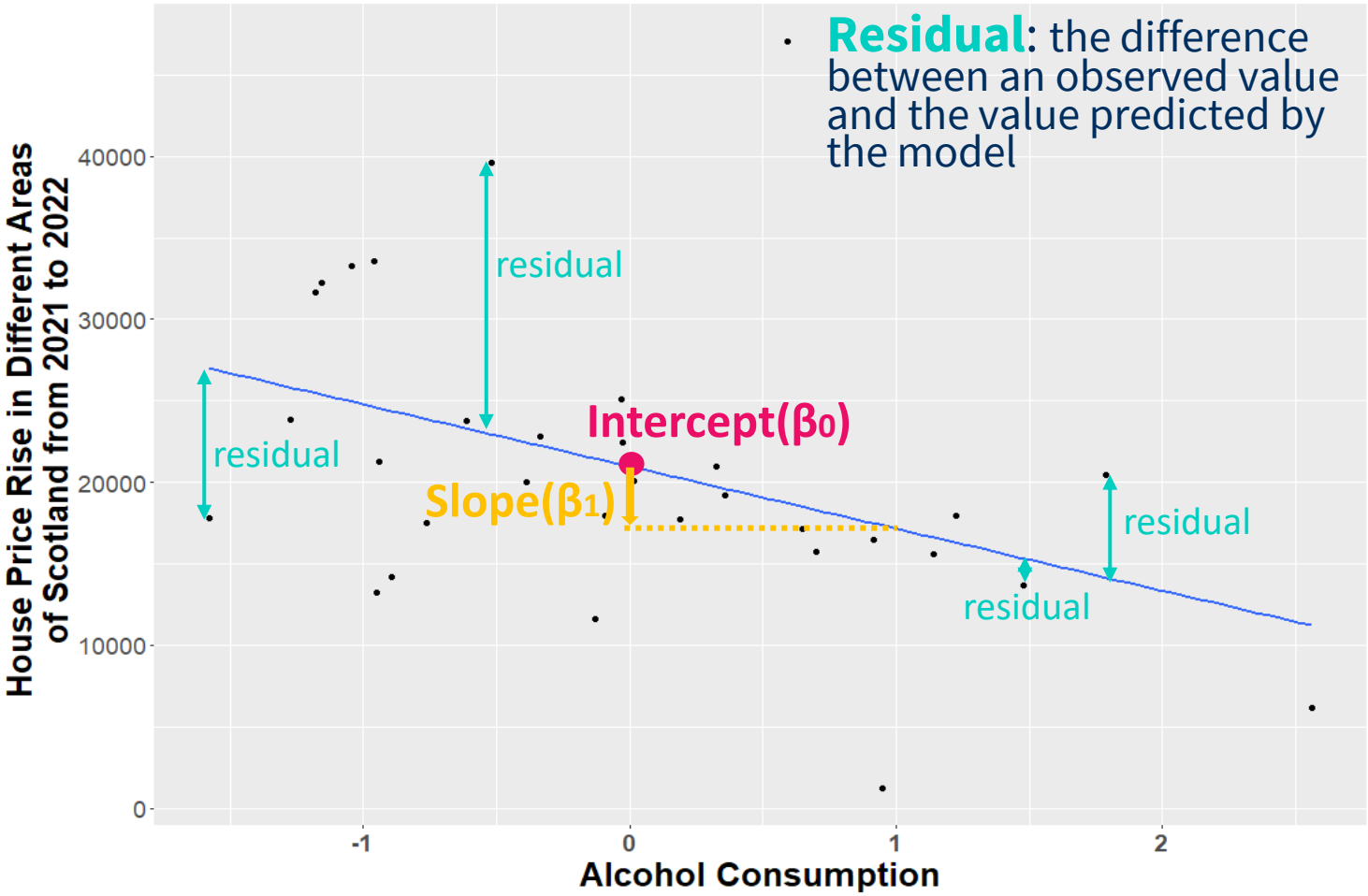
How good is our model?

- **Model evaluation**
- How much variance does the model account for?
- How much does the model fit improve over chance?



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Model Comparison & Selection

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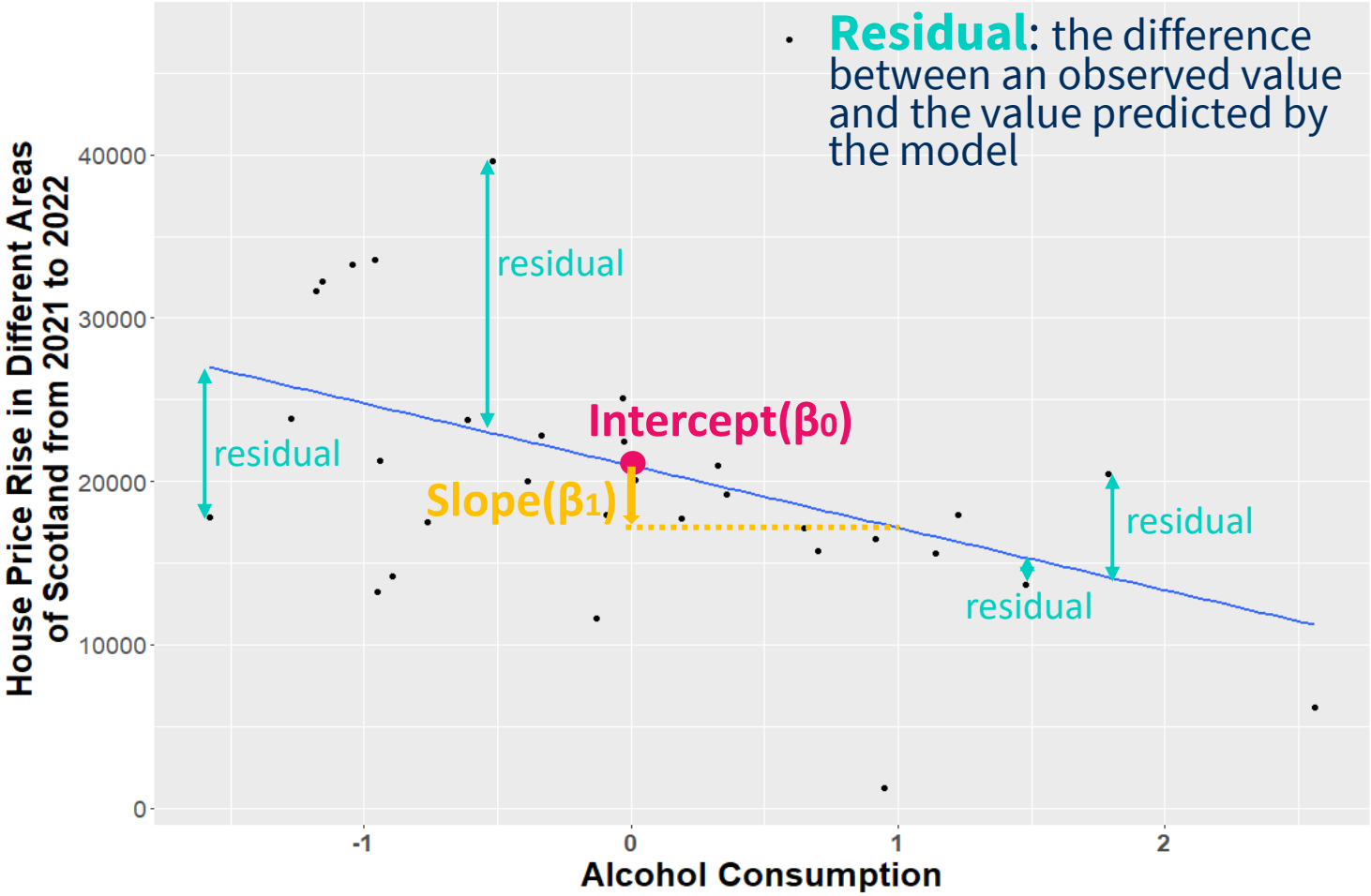
- Fit more than one model
- Compare goodness of fit
- Test significance

Does model fit improve significantly if we add an additional predictor / independent variable ?

(in R: **anova(**m1, m2**)**)



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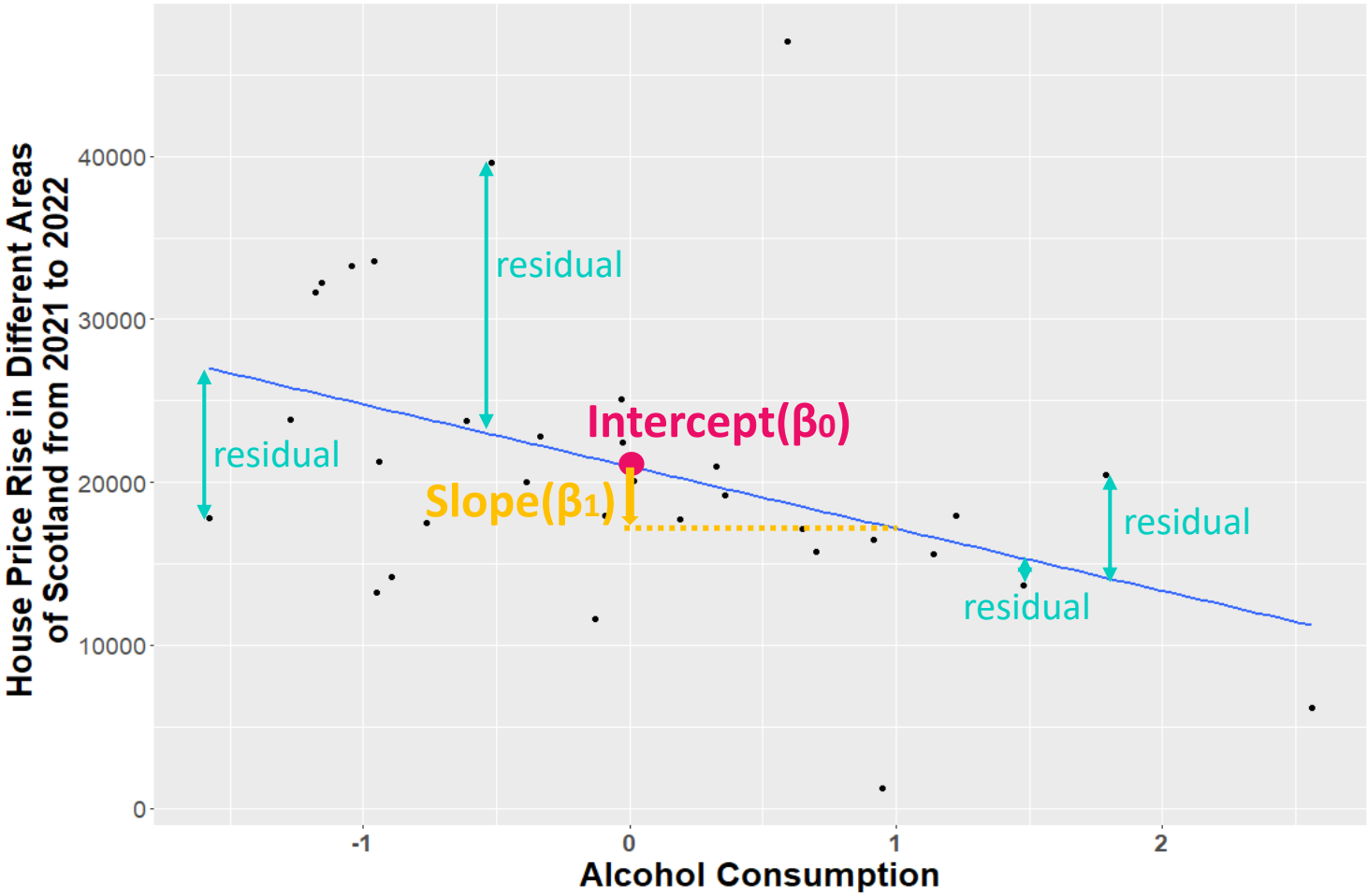
Model Assumption Checks

- **Nature of the model**

- the relation between predictor and outcome has to be linear

- **Nature of the errors (i.e., residuals)**

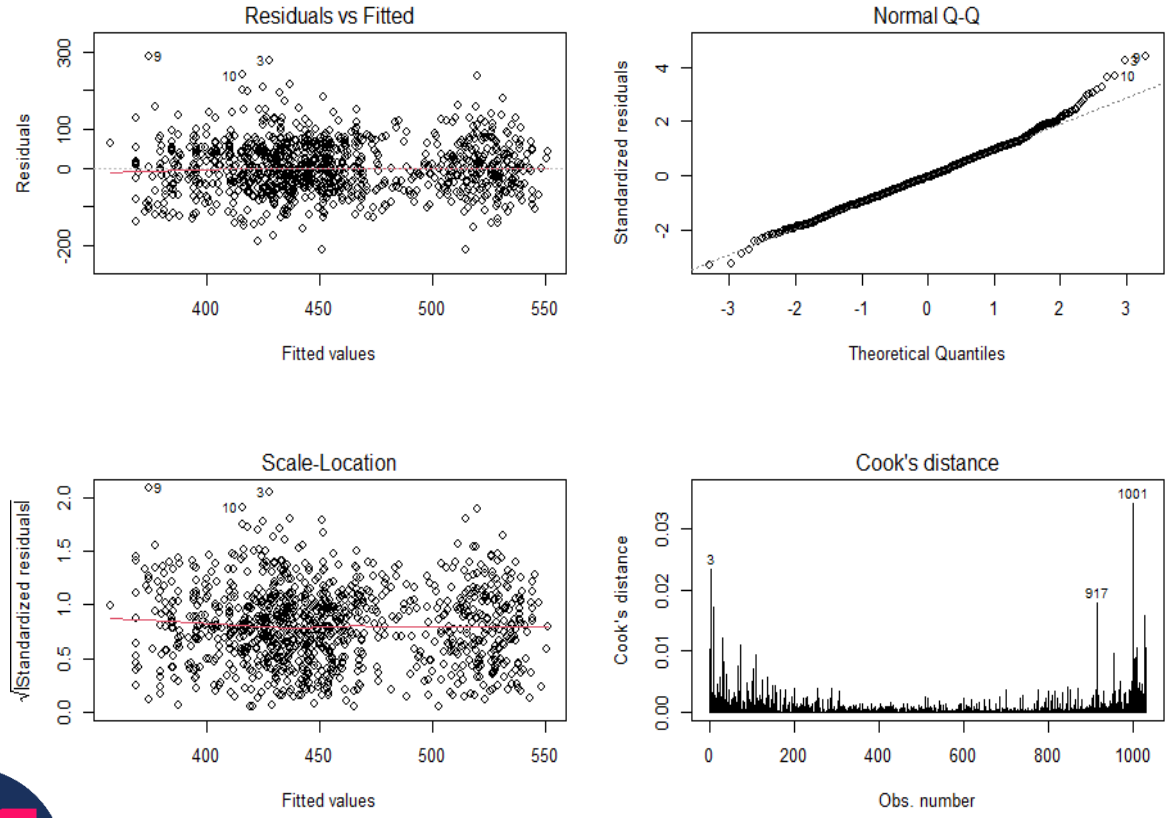
- normal and independent of each other



ASSUMPTION CHECK

- Required(**LINE**):
  - **L**inearity of relationships
  - **Independence of residuals**
  - **N**ormality of residuals
  - **E**qual variances for residuals

- Desirable:
  - uncorrelated predictors (no collinearity) for multiple regression
  - no outliers



Time for a Break

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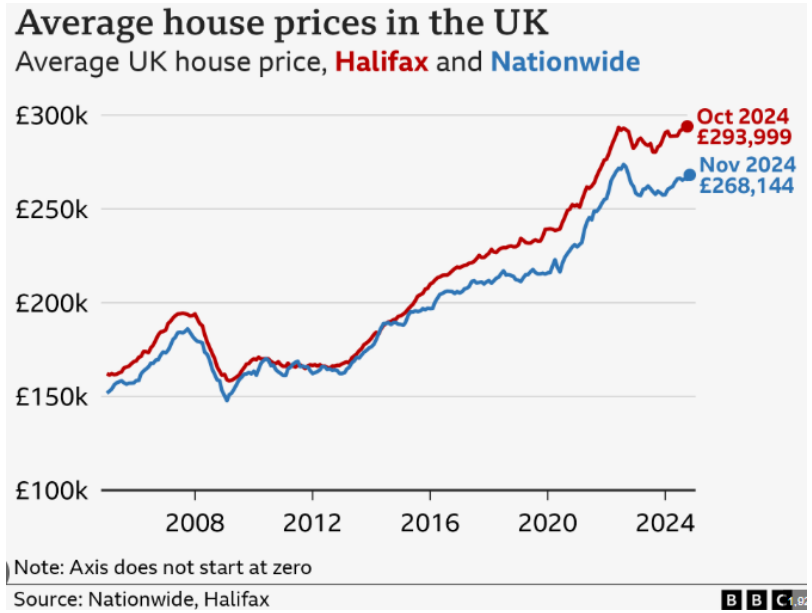
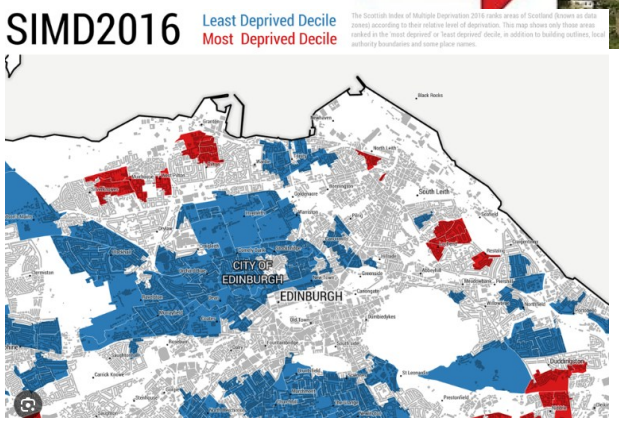
Regression in the real world





Is housing price increase associated with

- energy bill ?
- alcohol consumption?
- location (does it differ in rural and urban areas) ?
- deprivation (indexed by SIMD score or SIMD category) ?

Let's find out!

Using R !

THANK YOU